

EXPLORATORY DATA ANALYSIS REPORT

CUSTOMER CHURN DATASET

STUDENT: AREMU FRIDAOUS EYITAYO

TOOLS USED: Python, Numpy, Pandas, Matplotlib, Seaborn

DATA OVERVIEW

Dataset Description:

The dataset used for this analysis contains customer information aimed at understanding factors influencing customer churn. The dataset contains [594,194] rows and [21] columns, comprising of both numeric and categorical variables.

Numeric variables include (tenure, MonthlyCharges, TotalCharges) while **categorical variables** include (gender, Partner, Dependent's, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, Churn).

KEY VISUALIZATION

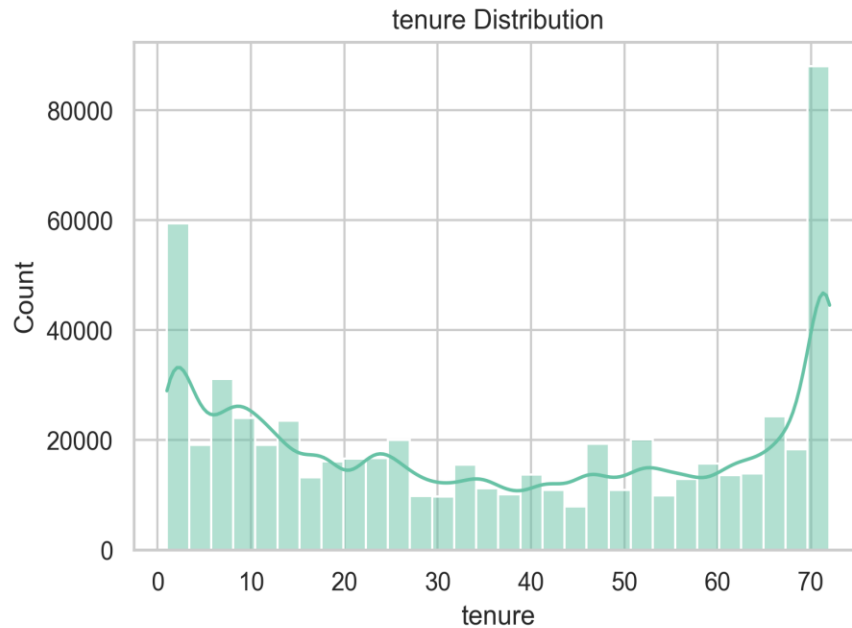
Several Visualizations were used to explore the dataset and understand patterns related to customer churn. These include boxplots, count plots and correlation heatmaps.

Boxplots were used to examine the relationship between numeric variables such as tenure, Monthly Charges against Churn. These plots helped to compare the distribution, median values and presence of outliers between customers who churned and those who did not.

Count plots were used to examine the relationship between categorical variables such as Contract, Partner against Churn. These visualizations provided insight into how different customer groups are distributed across churn categories.

Correlation heatmap was used to examine relationships between numeric variables. This helped identify strong and weak correlations, particularly the relationship between tenure, MonthlyCharges and TotalCharges.

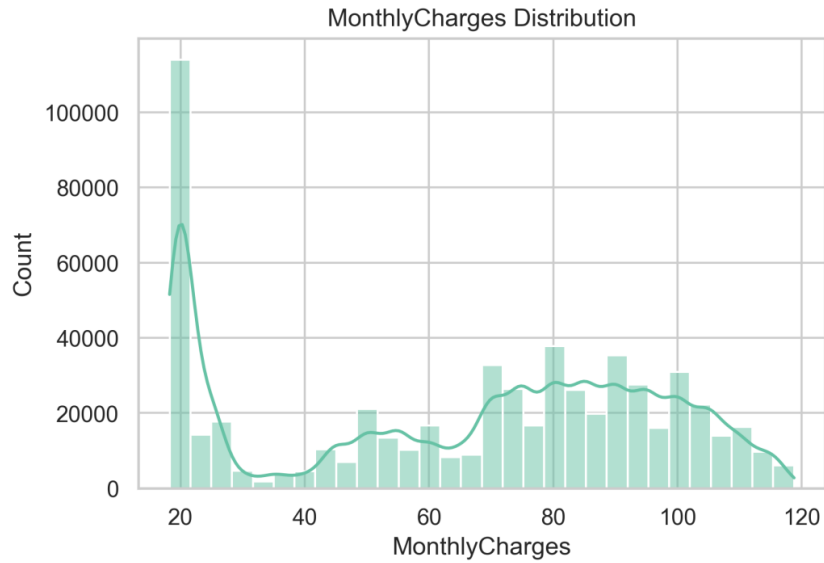
Figure 1: Tenure Distribution using Histogram



- A tall bar at the beginning(0-10months): This implies that a large number of customers are new or recently joined.
- Values drop in the middle(10-60months): This implies that not many customers stay in the middle range.
- Big spike at the end(~70-72months): This implies that many customers have stayed for a very long time (loyal customers).

The distribution of tenure shows a bimodal pattern, fewer customers fall within the midrange, indicating that customers tend to either leave early or stay for a long period

Figure 2: MonthlyCharges Distribution using Histogram



- There is a very tall bar at the beginning (~\$20/month): This implies that a large number of customers are on a cheap/basic plans
- The pike drops sharply and then rises again gradually from (~\$40-\$100)
- The middle (~\$30-\$40): This implies that very few customers pay mid-range prices
- A premium group spread across \$60-\$100 using more services.

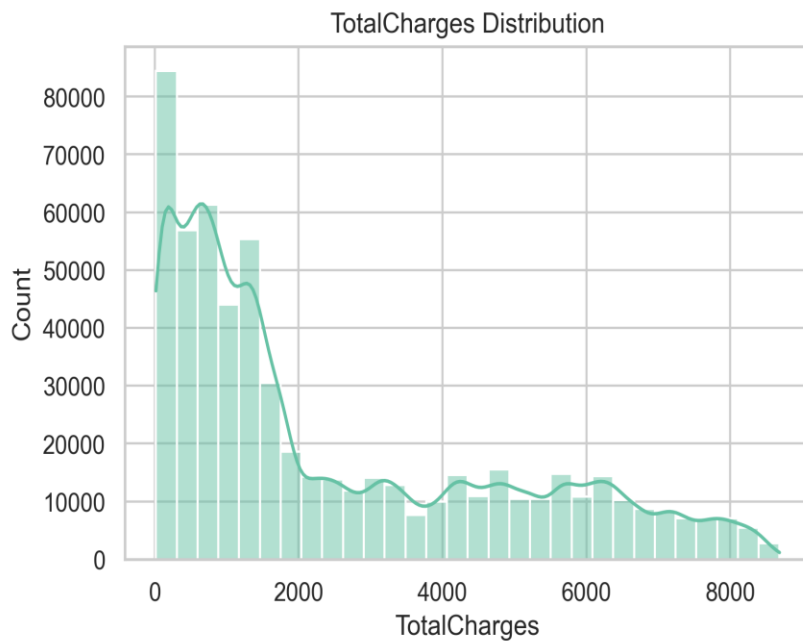
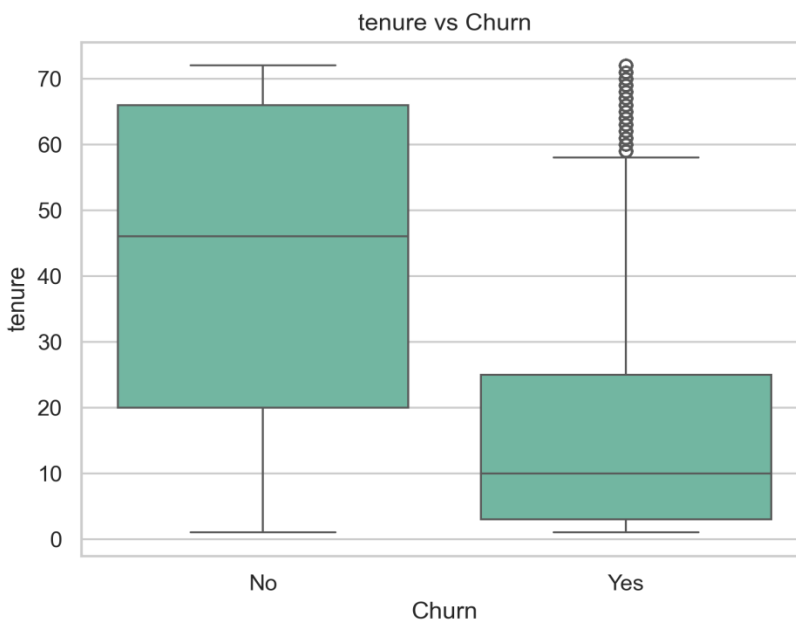


Figure 3: TotalCharges Distribution using Histogram

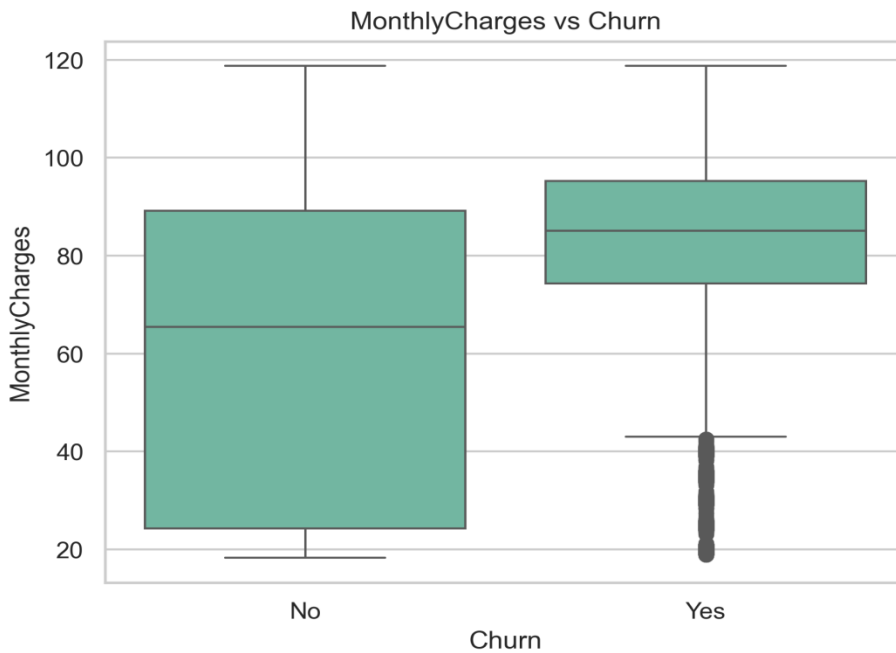
- There is a very tall bars at the beginning near \$0: This implies that a large number of customers have very low total charges. Why? These are likely new customers(low tenure), they haven't spent much yet
- The bar drops and then gradually increases again toward higher values (\$2000-\$8000): This represents long-term customers who have accumulated more charges over time.
- This pattern suggests that TotalCharges increases with customer tenure, indicating that long-term customers contribute significantly to total revenue.

Figure 4: Tenure vs Churn using Box plot



The boxplots shows that customers who do not churn have a wider distribution of tenure with a higher median, indicating they stay longer. In contrast, customers who churn have a lower median tenure and a smaller spread, meaning most of them leave early. There are also a few outliers among churned customers showing that some long-term customers still churn.

Figure 5: MonthlyCharges vs Churn using Boxplot



- The boxplot shows that customers who stayed pays cheaper prices, while churned customers tend to have higher monthly charges. This suggests that higher charges may increase the likelihood of churn.
- There are also outliers among churned customers, indicating that customers paying cheaper prices still left.

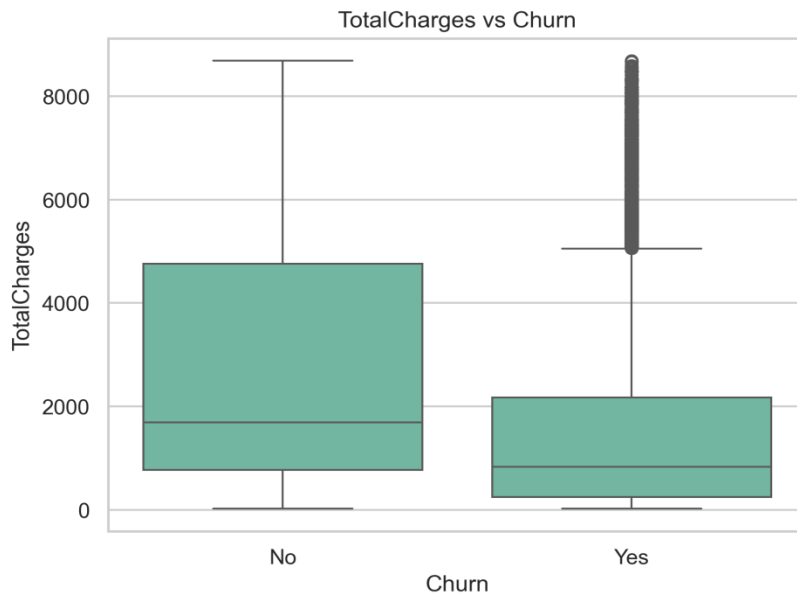


Figure 6: TotalCharges vs Churn using Boxplot

- The boxplot shows that customers who do not churn have a higher median total charge and a wider distribution, indicating they have stayed longer and accumulated more charges. In contrast, customers who churn have a lower median total charge and a smaller spread, showing they have lower total charges due to shorter tenure.
- Some outliers show that a few customers with high total charges still churn.

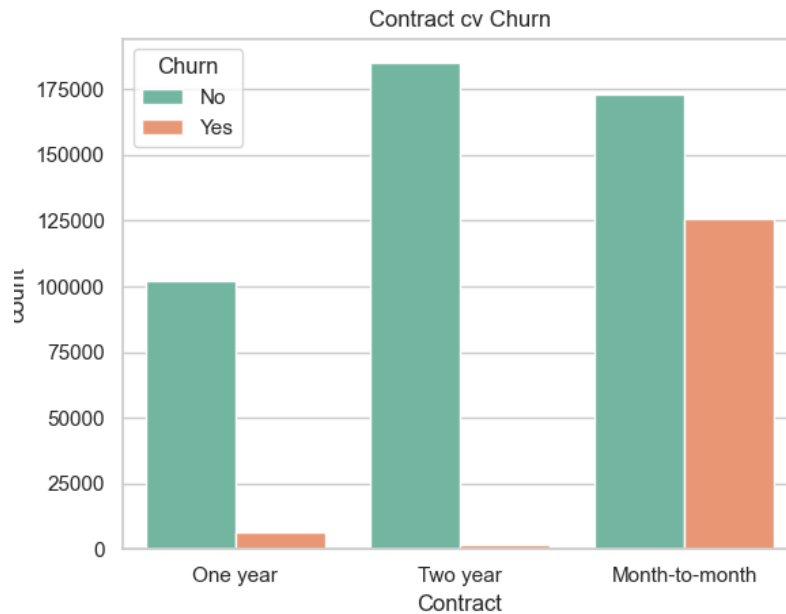


Figure 7: Contract vs Churn using Count plot

- The count plot shows that customers on month-to-month contracts have the highest churn rate, indicating that customers with no long term commitment can easily walk away from the service.
- Customers with one-year and two-year contracts are less likely to churn, suggesting that longer contract commitment help reduce customer churn.

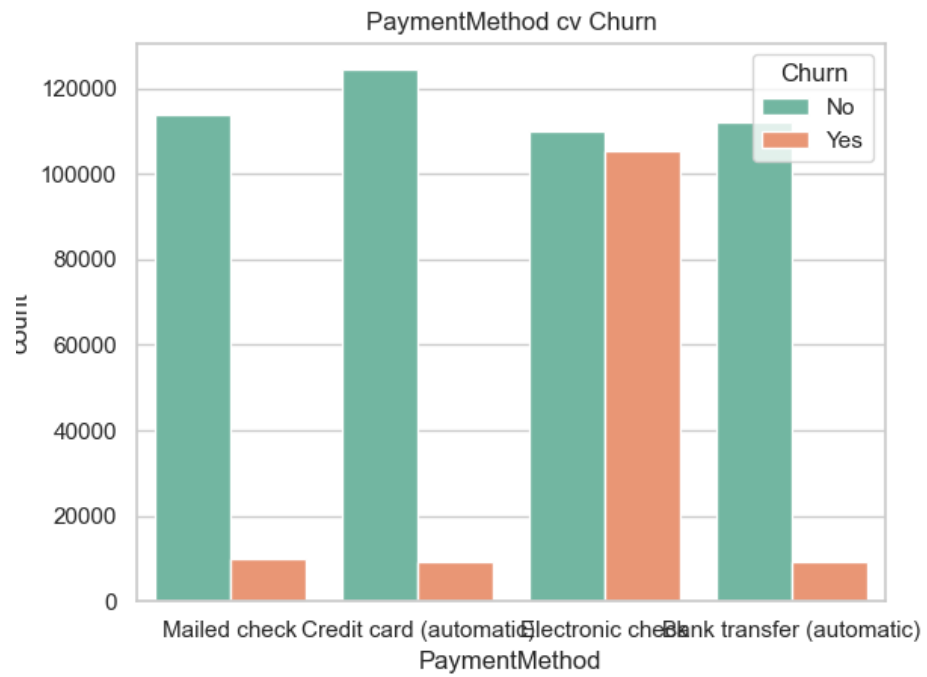


Figure 8: PaymentMethod vs Churn using Count plot

This count plot shows that customers using electronic check have the highest churn rate compared to other payment methods. This suggests that the type of payment method may influence the customer retention.

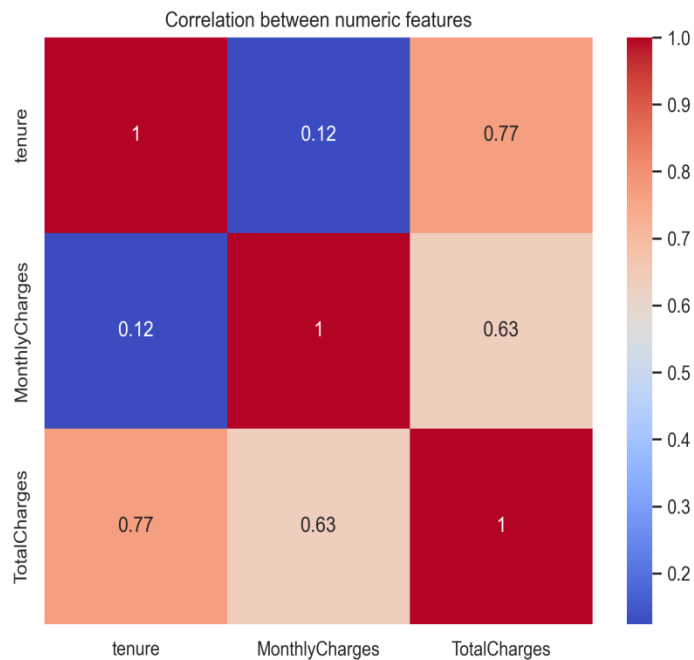


Figure 9: Correlation between Numeric Feature

- The correlation heatmap shows a strong positive relationship between tenure and TotalCharges, indicating that customers who stay longer tend to accumulate higher total charges.
- There is also a moderate positive relationship between MonthlyCharges and TotalCharges, meaning higher monthly payments contribute to higher total charges over time.
- However, tenure and MonthlyCharges show a weak correlation, suggesting that the amount a customer pays monthly is not strongly related to how long they stay.

MAJOR INSIGHTS

Insight 1: Churn Happens Early

Customers who churn usually have low tenure, while those who stay have much higher tenure. This shows that most customers leave within the first few months, making the early period the most critical.

Insight 2: High Monthly Charges Drive Churn

Customers who churn tend to pay higher monthly charges compared to those who stay. This suggests that higher pricing may lead to customer dissatisfaction and increase churn.

Insight 3: Contract Type is a Strong Predictor

Customers on month-to-month contracts churn more than those on one-year or two-year contracts. Longer contracts help retain customers and reduce churn.

Insight 4: Fiber Optic Customers Are at Risk

Customers using fiber optic services tend to churn more than those using other services like DSL. This may be due to higher costs or service expectations.

Insight 5: Two Customer Segments Exist

There are two main groups of customers: low-paying customers with short tenure who are more likely to churn, and higher-paying customers who stay longer but may still churn due to high costs.

OBSERVATIONS ABOUT CHURN BEHAVIOR

1. The data shows that customers with very low tenure(around 0-10months) have the highest churn rate. This indicates that many customers leave shortly after joining, making the early stage the most critical period for retention.
2. Customers who churn tend to pay higher monthly charges compared to those who stay. This suggests that customers are sensitive to pricing, and higher costs may lead to dissatisfaction and eventual churn.
3. There is a noticeable group of customers with very high tenure(around 70+ months) who do not churn. This shows that once customers stay for a long time, they are less likely to leave.
4. Customers on month-to-month contracts have the highest churn rate, while those on longer contracts(one-year or two-year) tend to stay. This indicates that longer contract commitments help reduce churn.